Certifying rare overflow risk of regional pickup-and-delivery plans

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Abstract

Medical, cold-chain and dangerous-goods distribution ask that a routing plan overflow its vehicles with probability at most γ, with γ between 10^{-3} and 10^{-6}. We study how to verify such a level for a given plan of the vehicle routing problem with simultaneous pickup and delivery. Under gamma demands with a Gaussian factor copula, the log asymptotics of a route's peak load have two end modes — total delivery at the start, total pickup at the end — with scales from secular equations. This anatomy explains when single-shift importance sampling fails: on routes whose two end scales are close, a single cross-entropy shift underestimates the overflow probability by about 5% in median (40% at the 90% quantile at 10^{-6}), and at 10^{-5} and below its relative variance stays near 5, so the error is not visible. Halving the pickups of other routes produces the same failure, so it follows the end-mode structure, not the region. We give a stage-mixture estimator placed at computed dominating points, with no pilot simulation, and extend it to a route-day simulator with stochastic travel and service times, where no exact method exists, and to hubs whose routes share demand factors, with a calibrated capacity search. On 148 instances built from population, postal and road data for Northern Vietnam and Guangdong, the route estimator agrees with an exact conditional dynamic program within 0.82% in every case where the program applies, and the route-day estimator agrees with 10^7-draw Monte Carlo within its error. The mixture reaches 1% precision in about three seconds per route, at least two orders of magnitude faster than the exact program, conditional Monte Carlo or subset simulation, while Glasserman–Li two-step sampling inherits the single-mode bias. Calibrated capacity buffers for hub overflow probabilities 10^{-3} and 10^{-5} are 35–43% and 64–76% (median by region), and treating routes as independent would overstate hub risk two- to threefold.

1 Introduction

Planning models for stochastic vehicle routing control risk through chance constraints, robust counterparts or distributionally robust ambiguity sets [Gounaris et al., 2013, Dinh et al., 2018, Ghosal and Wiesemann, 2020]; see Gendreau et al. [2016] for a review. They evaluate route risk inside the optimiser at moderate levels, with exact formulas, approximations or sampling. Applications with strict service levels need a step after optimisation: an estimate, with a confidence interval, showing that the plan actually produced meets $\mathbb{P}(\text{overflow}) \le \gamma$ for $\gamma \in [10^{-6}, 10^{-3}]$, for every route and for the plan as a whole, at the scale of a regional network.

At such levels crude Monte Carlo needs $1.96^2/(\varepsilon^2\gamma)$ draws for relative error ε at 95% confidence — 3.8×10^8 draws per route at $\gamma = 10^{-6}$ and $\varepsilon = 0.1$. Importance sampling (IS) is the standard remedy, and mean shifts tuned by the cross-entropy (CE) method are the practitioner's default [de Boer et al., 2005]. The failure of a pickup-and-delivery route is a union of events — the vehicle can overflow after any stop — and single-mode IS can then fail without warning [Glasserman and

Wang, 1997]. Mixtures over failure modes [Owen et al., 2019, Kurtz and Song, 2013, Geyer et al., 2019] cure this when the modes are known, at the price of pilot simulation when they are not.

Where IS is needed. For the demand-only route model used in planning there is a deterministic alternative: given the copula factors the demands are independent, so the overflow probability is a two-dimensional integral of a one-dimensional dynamic program. We implement it, call it the quadrature reference (it is discretised, extrapolated and truncated, so not exact), and use it both as the reference and as a competitor (Section 7.3). It needs no tuning, but at the grid that gives about 1% accuracy it takes [X] seconds per route and level, against [Y] seconds for the mixture estimator at the same precision. It does not extend to route days with stochastic travel and service times and lateness, where the event couples loads and times; there, of the methods we compare, only the mixture estimators are accurate, and the demand-only model is the setting in which their construction can be checked against a deterministic answer.

Contributions. (i) We formulate the certification of a given plan at service levels 10^{-3}–10^{-6}, route-wise and hub-wise, as a post-optimisation step. (ii) We derive the logarithmic asymptotics of the peak load of a pickup-and-delivery route under gamma marginals and a one-factor Gaussian copula: two end modes with spectral scales (Theorem 2, proof sketch). (iii) We show when single-shift IS loses logarithmic efficiency (Theorem 3) and measure its finite-level bias on real routes: the asymptotic threshold ($\varrho > 1/4$) holds for every route of our plans, but material bias appears only for $\varrho > 0.75$, follows the end-mode structure when the pickups are rescaled, is of the order of the computed missed-mode mass, and is invisible in the reported standard error at $\gamma \leq 10^{-5}$. (iv) We give a pilot-free stage-mixture estimator (SMIS) at computed dominating points, its extension to a route-day simulator without exact solution (MDIS), and a hub estimator for routes with shared factors, with attribution of hub risk to routes and a calibrated capacity search. (v) We evaluate on 148 region-scale instances against a quadrature reference, Glasserman–Li two-step IS, conditional Monte Carlo, subset simulation and CE. The IS ingredients are known — dominating points [Sadowsky and Bucklew, 1990], multiple-path failure of single shifts [Glasserman and Wang, 1997], mixtures as subsolutions [Dupuis and Wang, 2007], factor-copula two-step schemes [Glasserman and Li, 2005, Bassamboo et al., 2008] — and we do not claim new IS theory. What is new is their form for the pickup-and-delivery peak, the measured failure regime, and a certification procedure for regional plans.

2 Related work

Risk in vehicle routing. Chance-constrained and robust models [Gounaris et al., 2013, Dinh et al., 2018, Ghosal and Wiesemann, 2020] handle route risk inside the optimisation with exact formulas, moment bounds or scenario sampling [Gendreau et al., 2016]. Hazardous-materials routing measures risk as accident probability times consequence along links [Erkut and Verter, 1998]. We found no work that certifies an already optimised routing plan at rare levels with IS, and none that treats the mid-route peak of the simultaneous pickup-and-delivery load [Dethloff, 2001] as a rare event. The closest IS precedent in logistics is stockout simulation in production–inventory systems [Glasserman and Liu, 1996].

Rare-event simulation. Efficiency notions and dominating points are surveyed by Asmussen and Glynn [2007]; Glasserman and Wang [1997] show that a single shift is inefficient when the rare set has several ways of occurring; Dupuis and Wang [2007] build efficient schemes from subsolutions, of which mixtures are instances; Owen et al. [2019] bound the variance of a mixture that samples each event of a union conditionally; CE with Gaussian mixtures handles multimodal failure regions

[Kurtz and Song, 2013, Geyer et al., 2019, Papaioannou et al., 2019]. For Gaussian factor copulas, Glasserman and Li [2005] combine a factor shift with conditional exponential twisting, and Bassamboo et al. [2008] treat extremal dependence; for quadratic losses of Gaussian inputs, Glasserman et al. [2000] use the same eigenstructure that gives our scales. Subset simulation [Au and Beck, 2001] and sequential Monte Carlo splitting [Cérou et al., 2012] need no mode information; we use subset simulation as the model-free rival.

3 Problem

A route visits customers $1, \dots, L$. The vehicle leaves the depot with all deliveries $d_1, \dots, d_L$ and collects pickups $p_1, \dots, p_L$. The load after stop k is $L_k = D_k + P_k$ with $D_k = \sum_{i>k} d_i$ and $P_k = \sum_{i \le k} p_i$ $(k = 0, \dots, L)$, the peak is $F = \max_k L_k$, and the route overflows a capacity c if $F > c$.

Demand model. Each d_i is gamma with coefficient of variation cv, shape $\kappa = 1/\mathrm{cv}^2$ and scale $\theta_i^d = \bar{d}_i/\kappa$, likewise p_i. Dependence is a one-factor Gaussian copula per demand type: $d_i = G_{d,i}^{-1}(\Phi(Z_{d,i}))$ with $Z_{d,i} = \sqrt{\rho}\, W_d + \sqrt{1-\rho}\, E_{d,i}$, and similarly for pickups with a factor W_p independent of W_d. The input $u = (W_d, W_p, E_d, E_p)$ is standard normal. All customers of a hub share the two factors; hubs have separate factors. The baseline is cv $= 0.3$ and $\rho = 0.6$; Section 7.4 varies both, the pickup-to-delivery ratio and the marginal family. The independence of W_d and W_p is load-bearing: it makes the two end blocks independent, which the theory below uses.

Definition 1 (Certificate)**.** Given γ, a relative precision ε and a confidence $1-\delta$ (we use $\delta = 0.05$), a certificate for a route or a hub is an unbiased estimate $\hat{p}$ of its overflow probability p whose central-limit confidence interval has half-width at most $\varepsilon\hat{p}$, together with the check $\hat{p}(1+\varepsilon) \le \gamma$; for a hub, also the attribution of p to its routes. The interval is asymptotic; Section 7.4 reports its empirical coverage.

4 Structure of the overflow event

For an index set S with gamma scales θ_S let $C_S = \rho \mathbf{1}\mathbf{1}^\top + (1-\rho)I$ and

$$\lambda(S) = \lambda_{\max}\left(\Theta_S^{1/2} C_S \Theta_S^{1/2}\right),$$

the largest root of $1 = \rho \sum_{i \in S} \theta_i/(\lambda - (1-\rho)\theta_i)$. Write $\lambda_k^d = \lambda(\{i > k\})$ for the delivery block of stage k, $\lambda_k^p = \lambda(\{i \le k\})$ for its pickup block, $\lambda_k = \max(\lambda_k^d, \lambda_k^p)$, $\lambda^\star = \max(\lambda_0^d, \lambda_L^p)$ and $\varrho = \min(\lambda_0^d, \lambda_L^p)/\lambda^\star \in (0, 1]$.

Proposition 1. $\max(D_0, P_L) \le F \le D_0 + P_L$.

Theorem 2 (proof sketch)**.** *As $c \to \infty$ with the route fixed and unbounded gamma marginals,* $\log \mathbb{P}(L_k > c) \sim -c/\lambda_k$ *and* $\log \mathbb{P}(F > c) \sim -c/\lambda^\star$.

The proof (Appendix A) is a sketch: the uniformity of the logarithmic remainder of the gamma quantile (Lemma A1) is not written out. Our simulator clips Φ at $1 - 10^{-15}$, so its demands are bounded and the theorem describes the unclipped model; the clipping changes each input's law on a set of probability 10^{-15}, so on a route with $2L \le 80$ inputs it moves an overflow probability by less than 10^{-13}, far below $\gamma \ge 10^{-6}$.

So, on the logarithmic scale, a route overflows at the start (delivery block) or at the end (pickup block). Middle stages have strictly smaller scales: the stage where the two partial blocks cross has

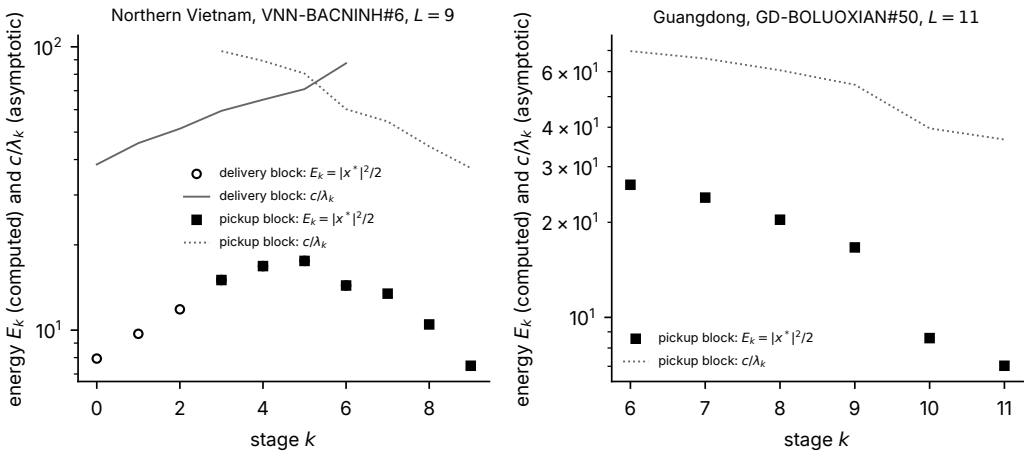


Figure 1: Computed dominating-point energies E_k of the stage events $\{L_k > c\}$ (open circles: delivery blocks; filled squares: pickup blocks) against the stage k, with the asymptotic prediction c/λ_k (lines), at $\gamma = 10^{-4}$. Left: a route of Northern Vietnam (balanced pickups and deliveries), where both end modes exist. Right: a route of Guangdong, where pickups are 2.7 times the deliveries and no delivery block reaches c (the demands are bounded by the clipping of Φ): the failure has a single end mode. The computed energies are 5–10 times below the asymptote, so the figure shows the pre-asymptotic gap as well as the anatomy.

$\lambda_k \le \lambda^\star/2$ on homogeneous routes. A stage k carries relative mass $\exp(-c(1/\lambda_k - 1/\lambda^\star)(1 + o(1)))$ (Proposition 4); with $c \approx \lambda^\star \log(1/\gamma)$ the weaker end mode carries $\gamma^{(1/\varrho-1)(1+o(1))}$. This is qualitative guidance only:

Remark 1 (Pre-asymptotic regime). For a three-stop test route with cv = 0.3, the exact dominating-point energy is 16% of its asymptotic value $c/\lambda^\star$ at $P \approx 9 \times 10^{-4}$ and 35% at $P \approx 1.6 \times 10^{-9}$; the logarithmic corrections of the gamma quantile are large at practical levels. The estimators therefore use computed dominating points, and the finite-level mass of a mode is measured by $\exp(-(E_m - E_{\min}))$ with exact energies E_m (Section 5).

Figure 1 shows the computed mode energies against the stage for one Northern Vietnam and one Guangdong route.

5 When a single shift fails

Mean-shift IS with shift τ uses the proposal $N(\tau, I)$ and weight $w(u) = \exp(-\langle u, \tau\rangle + |\tau|^2/2)$.

Theorem 3. *Let $\tau^\star$ be the asymptotic dominating point of the dominant end mode. (a) If $\varrho > 1/4$, the shift $\tau^\star$ is not logarithmically efficient:* $\log m_2(\tau^\star) - 2\log \mathbb{P}(F > c) \ge c\,(4/\lambda^\star - 1/(\varrho\lambda^\star))(1 + o(1)) > 0$. *(b) (proof sketch) If $\varrho \le 1/4$ and the dominant end block has second eigenvalue at most $\lambda^\star/4$, the shift is logarithmically efficient.*

Part (a) rests on Theorem 2; (a) and (b) have different hypotheses, so 1/4 is not a sharp threshold. The extra second moment in (a) comes from a region the proposal almost never visits, so the sample variance does not reveal it, and at finite levels the estimate is biased low by about the mass of the missed mode.

What the data show. The asymptotic criterion is uninformative for our plans: all 5 920 routes with two or more customers have $\varrho > 1/4$ (Guangdong median 0.36, Northern Vietnam median 0.93). The practical regime is narrower. On 120 routes (Table 2) the single CE shift has median error 0.14–0.23% in the ϱ bands with more than one route and $\varrho \le 0.75$, close to the mixture estimators, and 2.9, 5.1, 6.0 and 6.1% (90% quantile 7.6, 21.7, 26.7, 39.9%) when $\varrho > 0.75$ at $\gamma = 10^{-3}, \dots, 10^{-6}$. On those routes the error is an underestimate in 62% of the rows at 10^{-3} and in 93–95% at $\gamma \le 10^{-4}$ (median signed error -4.3 to -4.7% at $\gamma \le 10^{-4}$). It is visible at 10^{-3} and 10^{-4}, where the single shift's relative variance is 512 and 87, and invisible at 10^{-5} and 10^{-6}, where it is 5.2 and 5.5 (reported relative standard error 0.2% at 10^6 draws, against a 6% bias).

The missed-mode mass predicts the error. Let $\hat{m} = \exp(-(E_{\text{other}} - E_{\min}))$ be the weight that the computed dominating-point energies give the weaker end mode relative to the dominant one. It is computed from the route data in about a second and needs no sampling. We computed it for all 480 route–level rows (Figure 2, right). In the 320 rows with $\hat{m} < 10^{-6}$, all Guangdong, the single-shift error is at the noise level (median absolute error 0.19%, maximum 1.0%, against two standard errors of 0.43%); two more rows lie in $[10^{-6}, 10^{-3})$. In the 158 rows with $\hat{m} \ge 10^{-3}$, all from Northern Vietnam, the error is negative in 81–93% of the rows and grows with $\hat{m}$: the median is -0.4% for $\hat{m} \in [10^{-3}, 10^{-2})$, -2.2% for $[10^{-2}, 0.1)$ and -9.9% for $\hat{m} \ge 0.1$, and the median ratio of the absolute error to $\hat{m}$ is 0.98, 0.81 and 0.58. By level it is 0.22 at 10^{-3}, where the single shift still reaches part of the missed mode, and 0.74–0.79 at 10^{-4} to 10^{-6}. So at practical levels the single shift misses about $0.8\,\hat{m}$ of the probability, and $\hat{m}$ is a usable a-priori diagnostic: it is large exactly on the routes with $\varrho > 0.75$.

Region, ϱ and the assumed pickup rate. In our instances all rows with $\hat{m} \ge 10^{-3}$ are Vietnamese, where the pickup rate was set equal to the delivery rate because no regional pickup statistic is published, so the table above cannot separate the end-mode structure from region and assumption. We therefore changed the structure while keeping the routes. We rescaled the pickups of 20 Vietnamese routes by 0.5 and by 2 (Figure 3): ϱ falls from 0.93 to 0.49–0.51, the median $\hat{m}$ to below 10^{-10}, and the single-shift error is at most 1.7% in all 80 rows (median 0.1–0.2%), against medians of 5.1% and 6.1% at 10^{-4} and 10^{-6} over the 40 Vietnamese routes unscaled. In the other direction we halved the pickups of 30 Guangdong routes at 10^{-5} and 10^{-6}: ϱ rises to a median of 0.73, 42 of the 60 rows have $\varrho > 0.7$ and 13 have $\hat{m} \ge 10^{-3}$, four of those have an error beyond 2% (two routes, -41 to -43% and -6 to -8%), and the median ratio of error to $\hat{m}$ on the 13 rows is 0.84. The failure follows the end-mode structure, not the region. Correlating the delivery and pickup factors ($r = \pm 0.5$ between W_d and W_p, at 10^{-4}, 10 Vietnamese and 10 Guangdong routes each) leaves the Guangdong routes unaffected ($\varrho \approx 0.35$, errors below 0.9%) and makes the Vietnamese failure worse and, for $r = -0.5$, silent: the single shift underestimates by 2 to 80% (median 24%) on all 10 routes, while its reported relative variance is 4.3–5.3; for $r = +0.5$ the error has both signs and the relative variance is 5 to 25 000. The CE mixture agrees with the SMIS reference within 1.1% in both cases, so the reference is not the problem. SMIS keeps its construction here, since its dominating points are computed on the simulator, but the theory of Section 4 no longer applies and we report only this agreement. ϱ is computed in closed form and the threshold 0.75 remains empirical; $\hat{m}$ is the sharper indicator.

6 Estimators

SMIS. Algorithm 1 computes, for every stage k and block, a minimum-norm point u_m of $\{L_k(u) \ge c\}$ by constrained local minimisation (SLSQP) started on the block's Perron direction, with the exact quantile transform. The set is not convex, so u_m is a computed local minimiser, not a certified

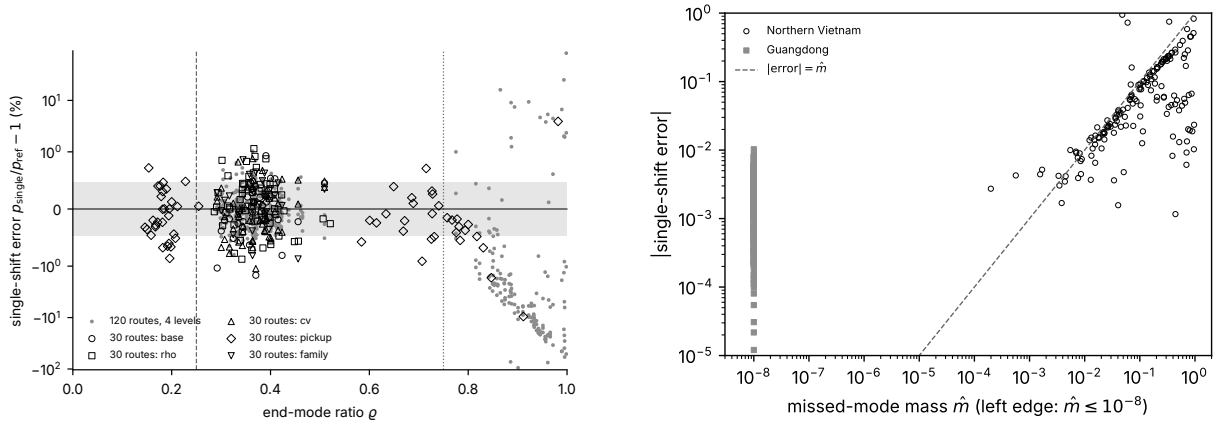


Figure 2: Left: signed relative error of the single CE shift against ϱ (grey dots: 120 routes at four levels; open markers: 30 Guangdong routes under the sensitivity settings at $\gamma = 10^{-4}$); symmetric-log scale, grey band ± 2 median standard errors; dashed and dotted lines at $\varrho = 1/4$ and 0.75. Right: absolute error of the single shift against the computed mass $\hat{m} = \exp(-(E_{\text{other}} - E_{\min}))$ of the weaker end mode, for all 480 route–level rows (circles: Northern Vietnam; squares: Guangdong, all at $\hat{m} \le 10^{-8}$, drawn at the left edge), with the line $|\text{error}| = \hat{m}$.

Algorithm 1 SMIS for one route

Require: route means $\bar{d}, \bar{p}$; cv, ρ; capacity c; η, a_0, N

1: **for** $k = 0, \dots, L$ and block $b \in \{d, p\}$ **do**
2: $v \leftarrow$ Perron vector of $\Theta_b^{1/2} C \Theta_b^{1/2}$ on the block's inputs
3: $u_{k,b} \leftarrow \arg\min\{|u|^2/2 : L_k(u) \ge c\}$ by SLSQP from the ray along v ▷ skip if c is unreachable
4: **end for**
5: keep modes with $E_m \le E_{\min} + \log(1/\eta)$; $a_m \leftarrow (1 - a_0) e^{-(E_m - E_{\min})} / \sum_{m'} e^{-(E_{m'} - E_{\min})}$
6: draw $u^{(1)}, \dots, u^{(N)} \sim q$; $\hat{p} \leftarrow N^{-1} \sum_j \mathbf{1}\{F(u^{(j)}) > c\}\, \varphi(u^{(j)})/q(u^{(j)})$
7: **return** $\hat{p}$, its standard error, ESS

global one. Modes with energy $E_m = |u_m|^2/2$ within $\log(1/\eta)$ of the smallest are kept, and the estimator samples from

$$q = a_0\, N(0, I) + \sum_m a_m\, N(u_m, I), \qquad a_m \propto e^{-(E_m - E_{\min})},$$

with likelihood ratio $\varphi(u)/q(u) = \left(a_0 + \sum_m a_m e^{\langle u, u_m \rangle - |u_m|^2/2}\right)^{-1}$. The tuning constants are $\eta = 10^{-3}$ (hubs 10^{-4}) and $a_0 = 0.05$, fixed for every experiment; there is no pilot simulation. An optional variant (SMIS3) adds 3 points on arcs between modes of adjacent stages; it changes nothing measurable on our routes and we report $n_{\text{arc}} = 0$ (SMIS0) throughout. Theorem 5 (Appendix) is a sketch of logarithmic efficiency under spectral-gap conditions; at $\rho = 0.6$ they hold for every end block of our plans, at $\rho = 0.3$ for 0.4% of the routes (1.4% of the end blocks), and Section 7.4 tests SMIS in both cases.

Cost: $2(L + 1)$ local minimisations of dimension $2 + 2L$, each a few dozen quantile evaluations, then N simulator calls and $N \times$ (#modes) inner products.

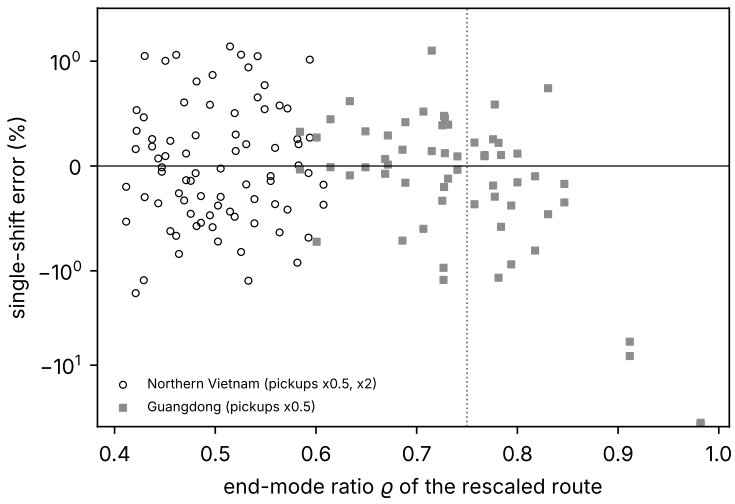


Figure 3: Single-shift error against the end-mode ratio ϱ of the rescaled route. Open circles: 20 Vietnamese routes with the pickups scaled by 0.5 and by 2, at 10^{-4} and 10^{-6} (80 rows; $\varrho \approx 0.5$). Squares: 30 Guangdong routes with the pickups scaled by 0.5, at 10^{-5} and 10^{-6} (60 rows). Reference: SMIS with 10^6 draws; symmetric-log scale; dotted line at $\varrho = 0.75$.

MDIS. The same construction applies to any simulator that exposes its failure modes as scalar functions $g_m(u) > 0$ of a Gaussian input. We use it for a route-day model (Section 7.5): demands as above, lognormal travel times per leg $T_l = \bar{t}_l \exp(\sigma_T(\sqrt{\rho_T}V + \sqrt{1-\rho_T}H_l) - \sigma_T^2/2)$ with $\bar{t}_l$ the road distance at 40 km/h, $\sigma_T = 0.25$ and a shared congestion factor ($\rho_T = 0.5$), service times of 10 minutes plus 6 minutes per tonne handled, and a failure if the vehicle overflows at any stage, arrives at any stop later than its nominal time plus a slack s, or returns later than nominal plus $2s$. Dominating points of the load, lateness and return modes are computed from several structured starts. No exact method applies: the event couples a maximum of copula sums with lognormal sums and load-dependent service.

Hubs. The routes of a hub share the factors, so the hub event $\bigcup_r \{F_r > c\}$ is again a union of stage events, now in dimension $2 + 2n$ for n customers. Every route's SMIS modes are embedded in the hub input (the two factors and that route's coordinates, zeros elsewhere), modes are kept over all routes with $\eta = 10^{-4}$, and the likelihood ratio

$$\log \frac{\varphi(x)}{q(x)} = -\log\Big(a_0 + \sum_m a_m \exp\big(\langle x, \theta_m\rangle - |\theta_m|^2/2\big)\Big)$$

is computed with sparse inner products, since θ_m has $2 + 2L_r$ nonzero coordinates. From the same draws we estimate each route's overflow probability and the probability that it is the only route to overflow — an attribution of hub risk to routes. A route that cannot reach c contributes no mode and zero probability.

Capacity calibration. To report the buffer a hub needs for γ, we search for c with $\hat{P}_{\text{hub}}(c) = \gamma$: a Laplace-type proxy $\log \sum_r e^{-E_r(c)} = \log \gamma$ gives a starting point c_0 (the proxy is not a bound: in a first run it set capacities at which the hub probability was 15 to 28 times below γ, at the 90% quantile and in median), then at most three secant steps on $\log \hat{P}_{\text{hub}}(c) - \log \gamma$ with 2×10^4 hub

Table 1: Instances and plans (sample-average gate). Utilisation is the larger of mean total delivery and mean total pickup over Q, per route.

region	instances	customers	routes	stops q25/50/75 (max)	Q (t)	pickup/delivery	utilisation (median)	ϱ (median)
Northern Vietnam	26	2 612	375	6 / 7 / 8 (15)	15	1.01	0.62	0.93
Guangdong	122	37 487	5 867	5 / 6 / 8 (40)	30	2.70	0.65	0.36

draws per evaluation, stopping when $|\log(\hat{P}_{\text{hub}}/\gamma)| < \log 1.25$; the final $\hat{P}_{\text{hub}}$ uses 10^5 draws. The buffer is $b = c/Q - 1$.

Quadrature reference. Given (W_d, W_p) the demands are independent, so $\mathbb{P}(F > c \mid W)$ is a one-dimensional dynamic program over the discretised load (we use $V = 256$ and 512 levels with Richardson extrapolation), and $\mathbb{P}(F > c) = \iint \mathbb{P}(F > c \mid w)\, \varphi(w_d)\varphi(w_p)\, dw$ is a product trapezoid on $[-6, 9]^2$. For a hub, the routes are conditionally independent, so $\mathbb{P}_{\text{hub}} = \iint [1 - \prod_r (1 - \mathbb{P}(F_r > c \mid w))]\varphi\varphi\, dw$. The error estimate combines the Richardson difference and a nested-grid quadrature difference.

7 Experiments

7.1 Instances and plans

We built 148 instances, one per depot area: the 25 provinces of Northern Vietnam (Hanoi split into two hubs) and 122 county-level areas of Guangdong (Table 1). Service points aggregate Kontur population hexagons (15 000 people per point in Vietnam, 4 000 in Guangdong) by weighted k-means; distances are OSRM road distances. Daily demand rates come from official postal statistics: 0.087 parcels per person per day in Vietnam (3.2 billion items in 2024) and, for Guangdong, 0.34 received and 0.91 sent per person per day (2024 volumes). The following are assumptions: 1 kg per parcel; truck capacities of 15 (Vietnam) and 30 (Guangdong) tonnes; the Vietnamese pickup rate equal to the delivery rate (no regional figure is published). The service points represent 35.3 million people in Northern Vietnam and 115.5 million in Guangdong (Kontur 2023 grid; the official Guangdong resident count for 2024 is 127.8 million, the difference being coastal hexagons outside the county polygons); Vietnamese provinces follow the pre-2025 boundaries. Plans are solved by ALNS with the sample-average chance gate; we certify these plans. Routes are short: the median route has 6 (Guangdong) and 7 (Vietnam) stops.

7.2 Routes under the demand model

We use 120 routes (80 Guangdong, 40 Northern Vietnam; 4 to 15 stops, median 7) and four levels $\gamma \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$; c is set so that the overflow probability is close to γ. Every estimator uses $N = 10^6$ draws. The CE mixture uses a pilot of 0.8–1.2×10^5 draws per route. Table 2 reports errors against the reference.

Against the quadrature reference. On the 89 routes with $L \le 8$ at $\gamma \in \{10^{-3}, 10^{-4}\}$ (178 cases), SMIS deviates from the quadrature reference by 0.15 and 0.16% in median, 0.38 and 0.44% at the 90% quantile, and at most 0.82%; the deviation is within 1.96 reported standard errors in 90% and 87% of cases (the program has its own discretisation error). The reference also covers $\gamma = 10^{-6}$ and routes with more than eight stops (Section 7.3): on the ten longest of 30 routes ($L \ge 9$) SMIS deviates by 0.13% (10^{-4}) and 0.20% (10^{-6}) in median and at most 0.58%.

Table 2: Route model: relative error against the reference (quadrature reference where available, else CE mixture), median / 90% quantile, and relative variance (median). 120 routes, $N = 10^6$; band $(0.5, 0.75]$ has one route and is omitted.

γ	ϱ band	n	SMIS	CE mixture	single shift	RV SMIS	RV single
10^{-3}	$[0, 1/3]$	17	0.19 / 0.58%	0.12 / 0.39%	0.17 / 0.45%	3.76	3.51
	$(1/3, 0.5]$	62	0.13 / 0.38%	0.09 / 0.30%	0.18 / 0.43%	3.78	3.51
	$(0.75, 1]$	40	0.22 / 0.44%	0.05 / 0.32%	2.86 / 7.57%	4.05	512
10^{-4}	$[0, 1/3]$	17	0.16 / 0.51%	0.17 / 0.52%	0.14 / 0.65%	4.50	4.23
	$(1/3, 0.5]$	62	0.17 / 0.45%	0.10 / 0.36%	0.18 / 0.44%	4.52	4.23
	$(0.75, 1]$	40	0.20 / 0.42%	0.07 / 0.36%	5.10 / 21.7%	4.71	87
10^{-5}	$[0, 1/3]$	17	0.19 / 0.55%	ref.	0.15 / 0.45%	5.18	4.88
	$(1/3, 0.5]$	62	0.22 / 0.45%	ref.	0.18 / 0.51%	5.18	4.88
	$(0.75, 1]$	40	0.26 / 0.58%	ref.	6.00 / 26.7%	5.31	5.22
10^{-6}	$[0, 1/3]$	17	0.24 / 0.61%	ref.	0.19 / 0.60%	5.79	5.46
	$(1/3, 0.5]$	62	0.26 / 0.55%	ref.	0.23 / 0.54%	5.79	5.46
	$(0.75, 1]$	40	0.22 / 0.52%	ref.	6.05 / 39.9%	5.88	5.50

Across all levels. SMIS has relative error at most 0.61% at the 90% quantile in every ϱ band with more than one route, relative variance 3.8–5.9 (CE mixture 3.8–6.0, single shift 3.5–5.5 where it is unbiased), 2–7 components, and a median build time of 0.39–0.86 s. At $\gamma \le 10^{-5}$ the reference for routes without an quadrature reference is the CE mixture.

Samples. Counting CE's pilot, the CE mixture needs 224–276 times more draws than SMIS for $\varepsilon = 0.1$ at 95% confidence and 3.2–3.8 times more for $\varepsilon = 0.01$: the advantage is the saved pilot, not a smaller variance.

7.3 Classical rivals and the quadrature reference

On 30 of the 120 routes (15 with $\varrho > 0.75$) at $\gamma = 10^{-4}$ and 10^{-6} we compare all methods with the quadrature reference (Table 3). The rivals are Glasserman–Li two-step IS [Glasserman and Li, 2005] with the factor shift at the dominant point (crude inner sampling, inner exponential shift, and a shift tuned on a pilot grid), conditional Monte Carlo over the two factors with the exact inner program and a tuned factor shift, and subset simulation [Au and Beck, 2001] with five replications of 2 000 draws per level. Costs are stated per unit of work (a simulator call, or one inner program for conditional Monte Carlo) and converted to the wall time needed for relative precision ε at 95% confidence, build and pilot included, on one core.

Four findings from Table 3. First, SMIS agrees with the quadrature reference within 0.58% on every route, including the 15 routes with $\varrho > 0.75$ at $\gamma = 10^{-6}$, which the comparison with the CE mixture in Section 7.2 could not check independently. Second, the factor-based rivals inherit the single-mode failure, because both shift the factors towards one dominating point. On routes with $\varrho > 0.75$, Glasserman–Li with a shifted inner step is off by 7–16% in median at 2×10^4 calls; Table 4 repeats this at 10^6 calls (the budget of the other estimators), where its sampling error falls to about 0.5% and the error stays at -15% (10^{-4}) and -6.6% (10^{-6}), so the error is bias. Conditional Monte Carlo's relative variance jumps from about 1 to 57–153 per inner program on the same routes, the same missed-mode effect seen through the variance. Third, subset simulation needs no mode information but has a relative variance of 700–4 400 per draw and needs 7–30 minutes for $\varepsilon = 0.01$. Fourth, the cost of the quadrature reference depends on its grid, and the grid we used as reference is finer than needed (next paragraph).

Table 3: Rivals against the exact conditional program, 30 routes (median over routes; error also as maximum). RV is per unit of work; $t(\varepsilon)$ is the one-core wall time for relative precision ε at 95% confidence, build and pilot included. The exact program needs no sampling; its time is per evaluation.

	$\gamma = 10^{-4}$			$\gamma = 10^{-6}$		
method	error med / max	RV	$t(0.01)$ (s)	error med / max	RV	$t(0.01)$ (s)
exact conditional program	reference	—	297	reference	—	303
SMIS	0.11 / 0.51%	4.6	2.6	0.16 / 0.58%	5.8	3.3
CE mixture	0.16 / 0.38%	4.6	4.6	0.29 / 0.85%	5.9	6.4
single CE shift	0.41 / 23.5%	4.3	2.5	0.36 / 94.7%	5.5	2.5
Glasserman–Li, crude inner	5.0 / 47.8%	29	15	7.1 / 48.9%	57	32
Glasserman–Li, shifted inner	3.1 / 47.1%	11	5.4	2.9 / 47.6%	13	6.6
Glasserman–Li, tuned shift	2.9 / 45.3%	83	41	4.1 / 82.7%	15	16
conditional Monte Carlo	2.6 / 93.7%	6.5	9 310	3.1 / 63.0%	1.4	2 190
subset simulation	8.1 / 38.1%	716	431	21.6 / 108%	4 430	1 800
routes with $\varrho > 0.75$ only (15): median error						
single CE shift	6.2%	212		9.7%	5.5	
Glasserman–Li, shifted inner	15.5%	12		7.4%	13	
conditional Monte Carlo	13.7%	57		13.2%	153	
SMIS	0.15%	4.7		0.15%	5.9	

Table 4: Budget-fair comparison, 30 routes, $N = 10^6$ simulator evaluations for each sampling method: absolute relative error against the quadrature reference (median / maximum) and relative variance per draw (median). SMIS, the two-end mixture and the CE mixture are unbiased on all routes; the Glasserman–Li estimator (shifted inner step) is biased low on balanced routes.

	$\gamma = 10^{-4}$		$\gamma = 10^{-6}$	
method	balanced (15)	other (15)	balanced (15)	other (15)
SMIS	0.15 / 0.57% (4.7)	0.17 / 0.38% (4.5)	0.12 / 0.57% (6.0)	0.24 / 0.58% (5.8)
two end modes only	0.13 / 0.59% (4.6)	0.08 / 0.45% (4.5)	0.13 / 0.39% (5.9)	0.19 / 0.60% (5.8)
CE mixture, pilot 2 000	0.10 / 0.30% (5.0)	0.12 / 0.56% (4.8)	0.13 / 0.51% (6.3)	0.23 / 0.49% (6.1)
Glasserman–Li, shifted inner	15.0 / 31.2% (30)	0.27 / 0.92% (10.6)	6.6 / 48.5% (13.9)	0.24 / 0.70% (12.2)

Budget-fair comparison. Table 4 gives every sampling method 10^6 simulator evaluations on the same 30 routes. Two further estimators separate what SMIS adds from what CE adds. The first keeps only the two end modes of the SMIS mixture (delivery block of stage 0, pickup block of stage L, with the same weights and defensive component); it is as accurate and as efficient as full SMIS (median error 0.08–0.19%, relative variance 4.5–5.9), so on these routes the modes at intermediate stages add nothing measurable and the gain over a single shift is the second end mode at its computed dominating point. The second is a CE mixture with a pilot of 2 000 draws per iteration; shrinking the iteration size did not shrink the pilot, which still takes 1.2–4.0×10^5 draws because the cross-entropy ladder runs for every stage, so the pilot saving of SMIS persists, while the accuracy of this CE variant equals SMIS (0.10–0.23%).

The quadrature reference at coarser grids. The reference of Table 3 uses 41^2 factor nodes and $V = 256$ and 512 levels. On the same 60 route–level rows, 41 nodes with $V = 128$ and 256 differs from it by 0.01% in median (0.44% at the 90% quantile, 0.97% at most) and costs about a third of the time (95 s against 300 s per evaluation under load). With 21 nodes the error is 1.2% in median (8.5% at most) and with 11 nodes 48%: the trapezoid rule on $[-6, 9]^2$ with a step above

Table 5: Sensitivity, 30 Guangdong routes, $\gamma = 10^{-4}$, 2×10^5 draws: absolute relative error (median / 90% quantile), median ϱ, and the capacity needed for 10^{-4} relative to the base setting (median).

setting	ϱ	SMIS	single shift	CE mixture	c/c_{base}
base ($\rho = 0.6$, cv 0.3, gamma)	0.37	0.27 / 0.77%	0.24 / 0.82%	0.26 / 0.82%	1
$\rho = 0.3$	0.37	0.34 / 0.78%	0.26 / 0.72%	0.28 / 0.60%	0.87
$\rho = 0.8$	0.37	0.33 / 0.95%	0.39 / 0.88%	0.39 / 0.85%	1.08
cv = 0.2	0.37	0.35 / 0.83%	0.26 / 0.67%	0.30 / 0.81%	0.79
cv = 0.5	0.37	0.37 / 0.66%	0.38 / 0.66%	0.35 / 0.66%	1.50
pickups $\times 0.5$	0.73	0.32 / 0.81%	0.34 / 0.99%	0.43 / 0.89%	0.50
pickups $\times 2$	0.18	0.40 / 0.77%	0.34 / 0.66%	0.28 / 0.61%	2.00
lognormal marginals	0.37	0.43 / 0.86%	0.29 / 0.67%	0.35 / 0.70%	1.09

0.5 does not resolve the tails that dominate the rare event. The cheapest grid that keeps the error below 1% is therefore 41 nodes with $V = 128$ and 256, and at that grid SMIS reaches the same 1% precision about [N] times faster (Table **??**). Section 7.6 compares the hub estimate with the exact hub program.

7.4 Sensitivity and coverage

On 30 Guangdong routes (4 to 14 stops) at $\gamma = 10^{-4}$ we vary one input at a time: $\rho \in \{0.3, 0.8\}$, cv $\in \{0.2, 0.5\}$, the pickup means scaled by $s \in \{0.5, 2\}$, and lognormal instead of gamma marginals with the same mean and cv (Table 5). In each setting the capacity is reset so that the overflow probability is close to 10^{-4}, every estimator uses 2×10^5 draws, and the reference is the exact program for gamma marginals with $L \leq 8$ (25 routes) and a 10^6-draw CE mixture otherwise.

SMIS stays within its sampling error (median 0.27–0.43% against a relative standard error of about 0.47%) in every setting, including $\rho = 0.3$, where the spectral-gap condition of Theorem 5 holds for only 0.4% of the routes of our plans (1.4% of the end blocks), and lognormal marginals, which the theory does not cover. The single shift is also accurate on these routes except where halving the pickups makes the two end scales nearly equal (Section 5). The last column is the model risk: the capacity needed for 10^{-4} moves by -21% and $+50\%$ when cv goes from 0.3 to 0.2 or 0.5, by -13% and $+8\%$ for ρ, and by $+9\%$ for lognormal marginals. These shifts are one to two orders of magnitude larger than the estimation error, so a certificate is only as good as the demand model, and cv in particular should be estimated from operator data.

Coverage. On 10 of the routes (base setting) we rebuild nothing and draw 300 independent replications of 10^4 draws each. The nominal 95% interval of SMIS covers the reference in 93.0–95.7% of replications (pooled 94.4%), with no zero estimates; the single shift, on these routes with $\varrho < 0.75$, covers in 94.0–98.0% (pooled 95.8%). At 10^4 draws the relative standard error is about 2%, so the CLT interval is adequate well below the sample sizes we use.

7.5 Route days without an exact method

On 60 routes we set the slack s and the capacity so that the load modes and the time modes each carry about half of the failure probability — a design that makes the failure multimodal on purpose, to test the estimator where a single shift is known to struggle. The threshold search is common to all methods and not counted. MDIS keeps 6–8 modes (median) of the 62–65 candidates (median); 16–21 candidate modes per route cannot reach their threshold and are dropped. Table 6 gives the results. At $\gamma = 10^{-3}$, against 10^7-draw crude Monte Carlo (relative standard error about 1%),

Table 6: Route-day simulator, 60 routes: deviation (median / 90% quantile), relative variance (median), build time (median).

γ	MDIS vs crude	CE vs crude	single vs crude	MDIS vs CE	MDIS vs single	RV MDIS / CE / single	build MDIS / CE (s)
10^{-3}	0.76 / 1.77%	0.89 / 1.75%	2.18 / 21.9%	0.20 / 0.47%	1.98 / 28.4%	4.33 / 4.41 / 742	3.1 / 43.1
10^{-4}	—	—	—	0.24 / 0.53%	3.42 / 63.3%	5.25 / 5.24 / 1430	3.5 / 49.4
10^{-5}	—	—	—	0.23 / 0.59%	9.25 / 97.6%	6.12 / 6.04 / 357	3.8 / 55.1

Table 7: Plans at their planned capacity Q: share of routes with overflow probability at most γ (route SMIS), and the overflow probability of a hub, by region (148 hubs, 6 242 routes). No hub passes at any γ.

	hubs	routes	share of routes with $P_{\text{route}}(Q) \le \gamma$			
region	median $P_{\text{hub}}(Q)$	median $P_{\text{route}}(Q)$	10^{-3}	10^{-4}	10^{-5}	10^{-6}
Northern Vietnam (26 hubs, 375 routes)	7.9%	2.2%	11.5%	9.9%	9.6%	9.1%
Guangdong (122 hubs, 5 867 routes)	10.8%	2.4%	12.2%	8.6%	6.7%	5.8%
all	10.2%	2.4%	12.2%	8.7%	6.8%	6.0%

MDIS deviates by 0.76% in median and 1.77% at the 90% quantile, CE by 0.89% and 1.75%, and the single CE shift by 2.2% and 22%. At 10^{-4} and 10^{-5} we also ran crude Monte Carlo, with 10^8 draws on 7 routes (1.0–1.2% relative standard error) and 2×10^8 draws on 3 routes (2.2%). MDIS deviates from it by 0.43% in median (2.1% at most) at 10^{-4} and by 2.6% (4.9% at most) at 10^{-5}, within about two standard errors of the reference in every case; the CE mixture is as close; the single CE shift is off by 0.4 to 38% at 10^{-4} and by 12 to 45% at 10^{-5}, underestimating on every route but one. MDIS and the CE mixture, built independently from the same simulator, agree within 0.24% in median and 0.59% at the 90% quantile over all 60 routes. MDIS is built in 3–4 s against 43–55 s for the CE pilot, and CE needs about 1 800 times more draws for $\varepsilon = 0.1$ and 18–19 times more for $\varepsilon = 0.01$.

7.6 Hubs

Do the plans pass? The plans were built with the sample-average gate at a per-route failure target of 0.10. We evaluated all 148 hubs and 6 242 routes at the planned capacity Q (hub SMIS with 10^5 draws, route SMIS with 2×10^4 draws per route). No hub meets any of the four levels (Table 7): the overflow probability of a hub at Q has a median of 7.9% in Northern Vietnam and 10.8% in Guangdong (10th to 90th percentile 6.2–9.9% and 8.1–13.8%). The median route overflows with probability 2.4% (90th percentile 3.8%), consistent with the target of 0.10 being met comfortably per route, and the shares of routes that pass are 12.2% at 10^{-3}, 8.7% at 10^{-4}, 6.8% at 10^{-5} and 6.0% at 10^{-6}. Most of the routes that pass are routes that cannot overflow at all: 4.7% of the routes have a largest possible load below Q. So a plan built for a per-route risk of 0.10 does not meet a service level of 10^{-3} or below anywhere; the certification step is what shows this, and the buffers below say how much capacity it takes to meet it.

Capacity needed. We certify 66 hubs: all 26 of Northern Vietnam and 40 of Guangdong, one drawn at random from each of 40 equal-size groups of the 122 hubs ordered by number of customers (25 to 597 customers, 5 to 100 routes). For each hub and $\gamma \in \{10^{-3}, 10^{-5}\}$ we search the common capacity c at which the hub overflow probability equals γ (Section 6) and report the buffer $b = c/Q - 1$ over the planned capacity (Table 8). All 132 searches converged, 121 of them after one

Table 8: Hubs: capacity buffer $b = c/Q - 1$ needed for hub overflow probability γ (median / 90% quantile over hubs), the buffer the Laplace-type proxy would set, and the region probability $1 - \prod_h (1 - P_h)$ over the certified hubs (Guangdong: the 40 sampled hubs only).

γ	region	hubs	buffer b	proxy buffer	P_{region}
10^{-3}	Northern Vietnam	26	35.1 / 37.7%	53.0%	0.024
	Guangdong	40	43.4 / 47.3%	68.5%	0.036
10^{-5}	Northern Vietnam	26	64.5 / 68.0%	81.5%	2.5×10^{-4}
	Guangdong	40	75.8 / 80.0%	96.8%	3.8×10^{-4}

secant step and 11 after two. Every route that can reach c has an SMIS mode; routes whose largest possible peak (with the demands bounded by the clipping of Φ) is at most c contribute exactly zero, which is why the exact hub program reports fewer reachable routes than the hub has (for example 3 of 5 at Bac Kan). The final estimate lies within a factor 1.25 of γ in every case (median $P_h/\gamma = 0.92$ at 10^{-3} and 0.94 at 10^{-5}, range 0.80–1.13). The final hub estimate takes a median of 36 s on one core (build 13 s plus 10^5 draws, 16–447 components, median 80); the whole task, with calibration, route-level baselines and checks, takes 3.4–4.4 minutes in median and at most 21 minutes.

The buffers are large because the plans were built with a sample-average gate at a moderate level and a coefficient of variation of 0.3 with strongly correlated demands ($\rho = 0.6$): moving from 10^{-3} to 10^{-5} costs another 30 points of capacity. A per-hub level of 10^{-3} gives, over the hubs we certified, a probability of 2.4% (26 hubs of Northern Vietnam) and 3.6% (40 of the 122 hubs of Guangdong) that some hub overflows on a given day; the same per-hub level over all 122 Guangdong hubs would give about 10%. A regional level γ therefore needs a per-hub level near γ divided by the number of hubs.

Checks. The hub estimate lies between $\max_r p_r$ and $\sum_r p_r$, with tolerance four combined standard errors, at all 132 points; the hub probability is 3.4–3.7 times the riskiest route's and 0.37–0.51 times the sum over routes (medians). Against 2×10^6-draw crude Monte Carlo on the 20 smallest hubs at $\gamma = 10^{-3}$ (relative standard error 2.2–2.5%), 19 of 20 standardised differences are within ± 2 (one is 2.84), but their mean is +0.59. The exact conditional program settles this on the six smallest hubs (25–55 customers, 5–9 routes): the hub estimate deviates from it by -0.8 to +2.0% (median +0.6%, five of six positive) with a relative standard error of 0.67%, so a small upward bias of the order of the standard error cannot be excluded; it is conservative for certification. The exact hub program takes 13–31 minutes per hub and level, against about 35 s for the hub estimate, and its cost grows linearly with the number of routes.

Shared factors. Treating the routes of a hub as independent overstates the hub probability by a factor 2.7 in median (4.6 at the 90% quantile) at 10^{-3} and 2.0 (2.9) at 10^{-5}, so a certificate built from route-wise estimates would demand too much capacity. With the shared factors, 47% (at 10^{-3}) and 33% (at 10^{-5}) of the overflow days of a hub have two or more routes overflowing (medians over hubs): standby capacity should be planned for clusters. The attribution shows how concentrated the risk is: the overflow probability of the riskiest route is 28–29% of the hub probability, and that of the three riskiest routes together 66–79% (medians over hubs; route events overlap, so these are not shares that sum to one; relative to the sum of the route probabilities they are 11–14% and 30–34%). Reinforcing a few routes, not all, is what a planner would do with the certificate.

7.7 A note on quantum amplitude estimation

Quantum amplitude estimation reduces the number of simulator calls for a probability quadratically [Brassard et al., 2002, Montanaro, 2015]. On the same 120 routes, counting oracle calls only (no qubit or error-correction cost) with explicit constants for a rigorous schedule, it needs 0.003–0.5 times the calls of crude Monte Carlo, but 27–3 550 times more than the sampling stage of classical IS (2.5–380 times more if the IS pilot is counted). With an IS-equipped oracle it needs fewer calls than classical IS only in an idealised accounting at $\varepsilon \leq 3\%$ (0.14–0.48 times). On this problem the mode structure, not the quadratic speed-up, is what makes certification cheap.

8 Conclusion

Rare overflow of a pickup-and-delivery route has a simple anatomy on the logarithmic scale — two end modes with closed-form scales — that tells when a single IS shift is at risk. In our plans the asymptotic criterion holds everywhere, while a material and silent bias appears only on routes with nearly balanced end scales. A mixture at computed dominating points removes it without pilot simulation, extends to route days without an exact method, and to hubs with shared factors. Against an exact conditional program that exists for the demand-only model, the mixture is about a hundred times faster at 1% precision and agrees within 0.6%; the factor-based rivals and subset simulation are slower or biased on balanced routes. On the regional plans, certified buffers are large (35–43% at 10^{-3}, 64–76% at 10^{-5}), a route-wise analysis that ignores the shared factors would overstate hub risk two- to threefold, and model risk in the demand variability moves the needed capacity far more than estimation error does. Open points: full proofs at the stated generality, region-wide factors that couple hubs, correlated pickup and delivery factors, and demand and travel models estimated from operator data.

Code and data. The instances, plans, simulator, estimators and every script that produces a number, table or figure in this paper are released with the paper. Table 1 is computed from the plan files; Table 2 from `run_rarecert_smis.py`; Table 3 and the exact hub checks from `run_rarecert_baselines.py`; Table 5 and the coverage study from `run_rarecert_sensitivity.py`; Table 6 from `run_rarecert_r2.py`; Table 8 from `run_rarecert_plan.py`; the figures from `make_rarecert_figures.py`. The long runs write per-task checkpoints and can be split into shards and merged.

A Proofs and sketches

Proof of Proposition 1. $L_0 = D_0$ and $L_L = P_L$ are stage loads, and $D_k \leq D_0$, $P_k \leq P_L$ for all k. □

Lemma A1 (quantile; stated without proof). For the gamma law, $x(z) = G^{-1}(\Phi(z)) = \theta z^2/2 + \theta(2\kappa - 1)\log z + O(1)$ as $z \to \infty$, and $0 \leq x(z) \leq x(0)$ for $z \leq 0$.

Proof sketch of Theorem 2. Upper bound: by Lemma A1, $D_k \leq \frac{1}{2} Z^\top \Theta Z + R$ with R a sum of logarithmic terms; for $t < 1/\lambda$, $\mathbb{E}\exp(t\,\frac{1}{2} Z^\top \Theta Z) = \det(I - t\Theta^{1/2} C \Theta^{1/2})^{-1/2} < \infty$, so Chernoff's bound gives the rate $1/\lambda$, and R is negligible if Lemma A1 holds uniformly (not written out). The delivery and pickup blocks have disjoint inputs, so their moment generating functions multiply and L_k has rate $1/\lambda_k$. Lower bound: a ball of radius $O(1)$ around $(1+\eta)\sqrt{2c/\lambda}\,u^\star$,

with $u^\star = B^\top \Theta^{1/2} v/\sqrt{\lambda}$ ($Z = Bu$, $BB^\top = C$, v the Perron vector), lies in the event. For F use Proposition 1.

Proposition 4 (from Theorem 2)**.** *For a fixed gap $\lambda_k < \lambda^\star$, $\mathbb{P}(L_k > c)/\mathbb{P}(F > c) = \exp(-c(1/\lambda_k - 1/\lambda^\star)(1 + o(1)))$.*

Lemma A2. Let $h(u) = \frac{1}{2} u^\top M u$ with $M = A^\top \Theta A$, $A \geq 0$, eigenvalues $\lambda_1 > \lambda_2$, $\kappa = \lambda_1/\lambda_2 \geq 4$, $h_+(u) = \frac{1}{2}|\Theta^{1/2}(Au)_+|^2$, and $\tau = r v_1$ with $r^2 = 2c/\lambda_1$. Then $h_+(u) \geq c$ implies $|u + \tau|^2 \geq 4|\tau|^2$. *Proof.* Write $u = a v_1 + u_\perp$, $b = |u_\perp|$. If $a \geq 0$, then $c \leq \frac{1}{2}(\lambda_1 a^2 + \lambda_2 b^2)$, so $b^2 \geq \kappa(r^2 - a^2)$ and $|u + \tau|^2 \geq (1 - \kappa)a^2 + 2ar + (1 + \kappa)r^2$, a concave function of a whose minimum on $[0, r]$ is $\min((1 + \kappa)r^2, 4r^2) = 4r^2$ for $\kappa \geq 3$; for $a \geq r$, $|u + \tau|^2 \geq (a + r)^2 \geq 4r^2$. If $a < 0$, then $Av_1 \geq 0$ gives $(Au)_+ \leq (Au_\perp)_+$, so $b^2 \geq \kappa r^2 \geq 4r^2$. □

Proof of Theorem 3(a), given Theorem 2. $\tau^\star$ acts on the dominant block's inputs only and the other end event depends on the other block's inputs only, so $m_2 \geq \mathbb{E}[w]\,\mathbb{P}(\text{other end}) = e^{|\tau^\star|^2}\mathbb{P}(\text{other end}) = \exp(2c/\lambda^\star - c/\lambda_{\text{other}} + o(c))$, while $2\log\mathbb{P}(F > c) = -2c/\lambda^\star(1 + o(1))$. □

Sketch of Theorem 3(b). Proposition 1 and Lemma A2 applied to the block-diagonal form of the two end blocks; the logarithmic remainders would be handled by Cauchy–Schwarz.

Theorem 5 (sketch)**.** *With a constant defensive weight, SMIS with modes at the asymptotic dominating points is logarithmically efficient if every retained block has $\kappa \geq 4$ and in every retained stage one block dominates the other by a factor of at least 4; stages with $\lambda_k \leq \lambda^\star/2$ need not be retained.*

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